
Ensoniq Performance Sampler External Command Specification

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MKB2

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1 Introduction and Overview

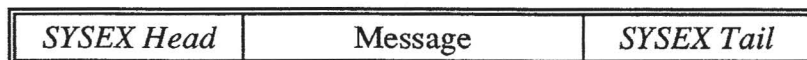
This manual describes the communication protocol and external commands used when the Ensoniq Performance Sampler (the EPS) is communicating with an external computer (the EXT). The protocol is designed to aid the implementation of editing programs running on EXT, and so this information is especially relevant to designers and programmers of editing programs.

The commands described here allow visual editing programs to collect information about instruments, layers, and wavesamples within the EPS, alter parameters, and process wavesample data. Since the primary function of a visual editor is viewing data, most existing editors spend a lot of time transferring data between EXT and EPS. Also, in many existing editors, one machine will process the data while the other waits for the processing machine before transferring the data to see the results. The data processing commands described in this manual save editing time while allowing EPS and EXT to independently process data and achieve the same results without the need for lots of data transfers. The GET WAVESAMPLE OVERVIEW command is designed to allow a visual editor running on a machine with few memory resources to view wavesample data without storing wavesample data in the editing machine.

The protocol can be used with MIDI System Exclusive or SCSI (Small Computer Systems Interface) messages. This manual contains the details for implementing this protocol using a MIDI interface. When a SCSI cable is connected between EXT and EPS, all data transfers to and from sound RAM (parameter and wavedata blocks) would occur using SCSI. This design will make the SCSI implementation easier when SCSI is available for the EPS.

2 MIDI System Exclusive Packet Pieces

A packet is a bunch of information, i.e. a message, in the form of a data stream. Each packet can be divided into three sections or pieces. The first and last packet pieces form the frame for a message. The message contains the commands described in section 4. Every message must be preceded with a SYSEX head and followed with a SYSEX tail. A complete packet looks like this:



2.1 MIDI System Exclusive Packet Head

This is the common system exclusive header which must be used on all SYSEX messages to and from the EPS. These four bytes are always sent preceding the message.

11110000	F0	System Exclusive status byte
00001111	0F	ENSONIQ Code
00000011	03	Ensoniq Performance Sampler ID Code
0000nnnn	0x	nnnn = x = Base MIDI channel number

2.2 MIDI System Exclusive Packet Tail

For every head there is a tail. This byte follows every message.

11110111	F7	End of System Exclusive
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2.3 Message Format

The EPS message format within the packet frame consists of a command byte followed by any number of data words. There are two word formats implemented in order to speed up message transfer by minimizing unnecessary data bytes.

The 12 bit format is used for data in most of the messages and consists of two 6 bit bytes. The 12 bits are right justified within a 16 bit register.

12 Bit Word Format	
00xxxxxx	x = Data word hi byte
00yyyyyy	y = Data word lo byte

This represents how the 12 least significant bits of the 16 bit word 0000xxxxxxxxyyyyyy would be transmitted.

The 16 bit word format, which consists of three MIDI bytes, is used when all 16 bits of a word need to be transmitted.

16 Bit Word Format	
0000HHHH	H = Hi 4 bits of data word
00hhhhhh	h = Middle 6 bits of data
00l11111	word
	l = Lo 6 bits of data word

This represents how the 16 bit word HHHHhhhhhh111111 would be transmitted.

3 Rules of the Game

These rules describe the communications protocol between the transmitter and the receiver. In general, for each command message sent by the transmitter, there is a response command sent by the receiver. Either device (EPS or EXT) can be transmitter and receiver simultaneously so that each one can be processing a command while waiting for a response from a command that it sent. However, a transmitter cannot send another command before receiving a response from a previous command except in WAIT mode which will be described in section 3.2.

3.1 General Command Exchange (NORMAL mode)

This set of rules describes how 99% of the commands should be processed. These rules allow an open or closed loop between the transmitter and receiver. An open loop exists when there is one cable from the transmitter to the receiver so that the transmitter will not receive response commands from the receiver. A closed loop uses two cables between the transmitter and receiver so that two-way communication is possible. A closed loop is more efficient since the transmitter can resume its operation immediately after receiving a response command rather than waiting for a timeout.

- The transmitter sends the message containing a command and parameters (except the response command) and starts the 2 second command timer. The transmitter should not send another command until it gets a response from the receiver or it times out. A few specific commands do not need responses and are noted in section 4, but the transmitter must still wait 2 seconds before sending another command.
- The receiver processes the command and sends a response command with the proper status code. The receiver should ignore all other commands while it is processing one or send a negative acknowledge (NAK) if another message is received during processing.
- If the transmitter times out, the link may be broken or it is an open loop. The transmitter must use its own discretion when processing a timeout error.

3.2 Special WAIT Command Exchange (WAIT mode)

This set of rules should be followed when the receiver sends a response command with the WAIT status code which initiates WAIT mode.

- The transmitter restarts the command timer with a 30 second value and sends a response command with the ACK status code, or it sends the CANCEL command. The receiver should wait for the transmitter's response before doing whatever it is that requires the WAIT mode.
- If the transmitter sent the CANCEL command, the receiver should abort the original command and return an ACK.
- If the transmitter sent an ACK, the receiver should process the original command and send the appropriate response command as soon as possible.
- If the receiver needs more time, it can send a response command with the WAIT status code again. The transmitter should re-enter wait mode by resetting the 30 second

command timer.

- If the transmitter times out, it must use its own discretion when handling the error.

An error message to the user or a longer wait are two possible timeout error responses.

4 Command List

This is the list of commands to be used in messages between EXT and EPS. These commands are the detailed description of the message which is enclosed in the SYSEX packet frame described in section 2. Unless otherwise specified, all command parameter words use the 12 bit word format as described in section 2.3.

4.1 General Communication Commands

Code and parameters	Name	Description
00	unused	not used
01 00 0a	RESPONSE	Send response to a received command. Status Code hi byte, (always 0) a = Status Code lo byte, see section 5 for a description of the Status Codes.
02	CANCEL	Abort current command processing. This is only valid in WAIT mode (it should be ignored in normal mode)

4.2 Parameter and Wavesample Data Transfer Commands

This set of commands are used to GET and PUT parameter values and transfer wavesample data. A GET command transmitted by EXT will prompt the EPS to transmit a PUT command. The EPS does not transmit GET commands.

NOTE on wavedata offsets: Wavedata offset address parameters are used internally as long words (20 bits). They are transmitted as two 12 bit words, e.g. the offset 00200 hex would be sent as the four parameter bytes 00, 00, 40, and 00 hex. Likewise, the offset 0AAAA hex would be sent as 00, 0A, 2A, and 2A hex.

NOTE: All parameter modification commands contain the current edit context defined by the edit instrument, layer, and wavesample.

03	GET INSTRUMENT	Request for instrument parameters.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
04	GET LAYER	Request for layer parameters.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
05	GET WAVESAMPLE PARAMETERS	Request for wavesample parameters.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]

06	GET WAVESAMPLE DATA	Request for wavesample data. See the example in section 8 of a data transfer between EXT and EPS.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
0e		Data range start offset (from beginning of wavesample data), Data offset = [0..FFFFFF]
0f		Start Offset = 0efgh (long word, 20 bits)
0g		see note at top of section 4.2
0h		Data range end offset (from beginning of wavesample data), Data offset = [0..FFFFFF]
0i		End Offset = 0ijkl (long word, 20 bits)
0j		see note at top of section 4.2
0k		
0l		
07	GET PITCH TABLE	Request for the pitch table data within the requested layer.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
08	GET PARAMETER	Request for a parameter value. See section 9 for the parameter numbers.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
0e		e = Parameter number hi byte
0f		f = Parameter number lo byte
09	unused	unused code

0A	GET WAVESAMPLE OVERVIEW	Request for wavesample data overview. This is never transmitted by EPS.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
0e		Data range start offset (from beginning of wavesample data), Data offset = [0..FFFFF]
0f		Start Offset = 0efgh (long word, 20 bits)
0h		see note at top of section 4.2
0i		Data range end offset (from beginning of wavesample data), Data offset = [0..FFFFF]
0j		End Offset = 0ijkl (long word, 20 bits)
0k		see note at top of section 4.2
0l		
0B	unused	unused code

Note on PUT Commands: All PUT commands which contain parameter blocks or wavedata are transmitted in two messages. The first message contains the command code and the edit context (instrument, layer, and wavesample). The second message is only the data. An ACK or a timeout after the first piece will cause the second piece to be sent as described in section 3.1. Refer to section 7 for the descriptions of the data in the second message.

NOTE: All parameter modification commands contain the current edit context defined by the edit instrument, layer, and wavesample.

0C	PUT INSTRUMENT	Instrument parameter block dump.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
		Instrument parameter data block as described in section 7.1 using the 16 bit word format described in section 2.3.

0D	PUT LAYER	Layer parameter block dump.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
		Layer parameter data block as described in section 7.2 using the 16 bit word format described in section 2.3.
0E	PUT WAVESAMPLE PARAMETERS	Wavesample parameter block dump.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
		Wavesample parameter data block as described in section 7.3 using the 16 bit word format described in section 2.3.
0F	PUT WAVESAMPLE DATA	Wavesample data block dump. See the example of a data transfer in section 8.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
0e		Data range start offset (from beginning of wavesample data), Data offset = [0..FFFFFF]
0f		Start Offset = 0efgh (long word, 20 bits)
0h		see note at top of section 4.2
0i		Data range end offset (from beginning of wavesample data), Data offset = [0..FFFFFF]
0j		End Offset = 0ijkl (long word, 20 bits)
0k		see note at top of section 4.2
0l		
		All 16 bit signed sample data words within the data range using the 16 bit word format described in section 2.3.

10	PUT PITCH TABLE	Transmit the pitch table data for the specified layer.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
		Pitch table block described in section 7.4 using the 16 bit word format described in section 2.3.
11	PUT PARAMETER	Update a parameter value for an instrument, layer, or wavesample. If this is a system parameter, the instrument, layer, and wavesample are irrelevant. See section 9 for the parameter numbers. NOTE: This does not need a response. A response is sent if the parameter number or value is erroneous.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
0e		e = Parameter number hi byte
0f		f = Parameter number lo byte, # = [0..255]
0g		g = Parameter value hi byte of hi word
0h		h = Parameter value lo byte of hi word
0i		i = Parameter value hi byte of lo word
0j		j = Parameter value lo byte of lo word

13	PUT WAVESAMPLE OVERVIEW	Wavesample data overview dump. NOTE: This is not received by the EPS.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
0e		Data range start offset (from beginning of wavesample data), Data offset = [0..FFFFFF]
0f		Start Offset = 0efgh (long word, 20 bits)
0g		see note at top of section 4.2
0h		Data range end offset (from beginning of wavesample data), Data offset = [0..FFFFFF]
0i		End Offset = 0ijkl (long word, 20 bits)
0j		see note at top of section 4.2
0k		
0l		
		512 data words. The data range is divided into 512 sections and the maximum 16 bit absolute sample value within each section is transmitted using the 16 bit word format described in 2.3.
14	unused	unused code

4.3 Instrument Editing Commands

This set of commands is used when editing an instrument and any of the objects within an instrument including layers and wavesamples.

NOTE: All instrument editing commands contain the current edit context of instrument, layer, and wavesample.

15	CREATE INSTRUMENT	Create a new instrument with no layers or wavesamples. The instrument must not already exist.
00		New instrument number hi byte (always 0)
0a		a = New instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
1C	DELETE INSTRUMENT	Delete the instrument and free up memory in the EPS. The instrument must already exist.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
12	COPY INSTRUMENT	Copy the source instrument to the destination instrument. The source instrument must already exist, and the destination must not already exist.
00		Source instrument number hi byte (always 0)
0a		a = Source instrument number lo byte, # = [0..7]
00		Source layer number hi byte (always 0)
0b		b = Source layer number lo byte, # = [0..7]
0c		c = Source wavesample number hi byte
0d		d = Source wavesample number lo byte, # = [1..127]
00		Destination instrument number hi byte (=0)
0e		e = Destination instrument number lo byte
00		Destination layer number hi byte (always 0)
0f		f = Destination layer number lo byte, # = [0..7]
00		Dummy destination wavesample hi byte.
00		Dummy destination wavesample lo byte.

16	CREATE LAYER	Define a new layer with one wavesample in an instrument. The new layer must not already exist.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		New layer number hi byte (always 0)
0b		b = New layer number lo byte, # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
17	DELETE LAYER	Undefine a layer in an instrument. All wavesamples in the layer are deleted too. The layer must already exist.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
18	COPY LAYER	Copy the source layer to the destination layer.
00		Source instrument number hi byte (always 0)
0a		a = Source instrument number lo byte, # = [0..7]
00		Source Layer number hi byte (always 0)
0b		b = Source Layer number lo byte, # = [0..7]
0c		c = Source wavesample number hi byte
0d		d = Source wavesample number lo byte, # = [1..127]
00		Destination instrument number hi byte (=0)
0e		e = Destination instrument number lo byte
00		Destination layer number hi byte (always 0)
0f		f = Destination layer number lo byte, # = [0..7]
00		Dummy destination wavesample hi byte.
00		Dummy destination wavesample lo byte.
00		Copy data flag hi byte. (always 0)
0g		g = Copy data flag lo byte. When set, data will be copied. clear = 00, set = 01

19	CREATE WAVE-SAMPLE	Create a new wavesample in a layer with a single cycle square wave for data. The wavesample must not already exist.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = New wavesample number hi byte
0d		d = New wavesample number lo byte, # = [1..127]
1A	DELETE WAVE-SAMPLE	Undefine a wavesample in a layer and free up RAM. The wavesample must already exist.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
1B	COPY WAVE-SAMPLE	Copy wavesample parameters from the source wavesample to a new wavesample on the destination layer. The wavedata is copied if the data copy flag is set.
00		Source instrument number hi byte (always 0)
0a		a = Source instrument number lo byte, # = [0..7]
00		Source layer number hi byte (always 0)
0b		b = Source layer number lo byte, # = [0..7]
0c		c = Source wavesample number hi byte
0d		d = Source wavesample number lo byte, # = [1..127]
00		Destination instrument number hi byte (=0)
0e		e = Destination instrument number lo byte
00		Destination layer number hi byte (always 0)
0f		f = Destination layer number lo byte, # = [0..7]
0g		g = Destination wavesample number hi byte
0h		h = Destination wavesample number lo byte
00		Copy data flag hi byte, (always 0)
0i		i = Copy data flag lo byte. When set, data will be copied. clear = 00, set = 01

1D	AUDITION WAVESAMPLES	Audition an old and new wavesample using the audition utilities of the EPS. After the command is executed, EPS will send a DELETE WAVESAMPLE command specifying the wavesample that was not kept. The EPS never transmits this command.
00		Old instrument number hi byte (always 0)
0a		a = Old instrument number lo byte, # = [0..7]
00		Old layer number hi byte (always 0)
0b		b = Old layer number lo byte, # = [0..7]
0c		c = Old wavesample number hi byte
0d		d = Old wavesample number lo byte, # = [1..127]
00		New instrument number hi byte (always 0)
0e		e = New instrument number lo byte, # = [0..7]
00		New layer number hi byte (always 0)
0f		f = New layer number lo byte, # = [0..7]
0g		g = New wavesample number hi byte
0h		h = New wavesample number lo byte, # = [1..127]
40	VIRTUAL BUTTON PRESS	Send a virtual button press. This should only be used if the desired operation cannot be done using the existing commands. See the list of valid button numbers in section 6
00		Button number hi byte. (always 0)
0a		a = Button number lo byte. # = [00..35]
44	CREATE PRESET	Create a preset with the current selection of instruments and patches. Refer to the user's manual for more information on presets.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
00		Preset number hi byte (always 0)
0e		e = Preset number lo byte, # = [0..7]

4.4 Wavesample Digital Sample Processing Commands

These commands alter wavesample data using the EPS Digital Sample Processing functions. See the EPS Digital Sample Processing specification for details on how the functions alter the data including the manipulation algorithms.

NOTE on wavedata offsets: See the note at the top of section 4.2 for a description of the use of wavedata offsets.

NOTE on scale factors: Scale factors are stored internally as a 16 bit word where the hi byte is an integer [0..255] and the lo byte is a 7 bit fraction [0..127]. The factor is transmitted using two words in the 12 bit format so that all 16 bits of the value can be sent. For example, scale factor 255.99 (hexadecimal FF7F) is transmitted by the four bytes 00, 0F, 3D, and 3F hex. Likewise, the scale factor 1.0 (hexadecimal 0100) is transmitted by 00, 00, 04, and 00. The 16 bit scale factor must be converted to two 12 bit numbers before being transmitted, and so they are similar to transmitting 20 bit data offsets using the 12 bit format.

NOTE on fade zones: The cross fade zone parameter must not be greater than the loop length nor greater than the length of data preceding the loop or following the loop. Thus, $FZ \leq (LE-LS)$, $FZ \leq (LS-WS)$, and $FZ \leq (WE-LE)$ where FZ is the cross fade zone, WS is the start of the wavedata, WE is the end of the wavedata, LS is the start of the loop, and LE is the end of the loop.

1E	TRUNCATE WAVESAMPLE	Delete any unused wavesample data before sample start and after sample end.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
1F	CLEAR WAVEDATA	Clear the wavesample data in the source range.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
0e		Data range start offset (from beginning of wavesample data), Data offset = [0..FFFFFF]
0f		
0g		Start Offset = 0efgh (long word, 20 bits)
0h		see note at top of section 4.2
0i		Data range end offset (from beginning of wavesample data), Data offset = [0..FFFFFF]
0j		
0k		End Offset = 0ijkl (long word, 20 bits)
0l		see note at top of section 4.2

20	COPY WAVEDATA	Copy the wavesample data from the source to the destination. The destination wavesample must already exist.
00		Source instrument number hi byte (always 0)
0a		a = Source instrument number lo byte, # = [0..7]
00		Source layer number hi byte (always 0)
0b		b = Source layer number lo byte, # = [0..7]
0c		c = Source wavesample number hi byte
0d		d = Source wavesample number lo byte, # = [1..127]
00		Destination instrument number hi byte (=0)
0e		e = Destination instrument number lo byte
00		Destination layer number hi byte (always 0)
0f		f = Destination layer number lo byte, # = [0..7]
0g		g = Destination wavesample number hi byte
0h		h = Destination wavesample number lo byte
0i		Source data range start offset (from beginning of source wavesample data),
0j		Data offset = [0..FFFFFF]
0k		Start Offset = 0ijkl (long word, 20 bits)
0l		Source data range end offset (from beginning of source wavesample data),
0m		Data offset = [0..FFFFFF]
0n		End Offset = 0mnop (long word, 20 bits)
0o		Destination data range start offset (from beginning of destination wavesample data),
0p		Data offset = [0..FFFFFF]
0q		Start Offset = 0qrst (long word, 20 bits)
0r		
0s		
0t		

21	ADD WAVEDATA	Add the source data to the destination and store the result in the destination.
00		Source instrument number hi byte (always 0)
0a		a = Source instrument number lo byte, # = [0..7]
00		Source layer number hi byte (always 0)
0b		b = Source layer number lo byte, # = [0..7]
0c		c = Source wavesample number hi byte
0d		d = Source wavesample number lo byte, # = [1..127]
00		Destination instrument number hi byte (=0)
0e		e = Destination instrument number lo byte
00		Destination layer number hi byte (always 0)
0f		f = Destination layer number lo byte, # = [0..7]
0g		g = Destination wavesample number hi byte
0h		h = Destination wavesample number lo byte
0i		Source data range start offset (from beginning of source wavesample data),
0j		Data offset = [0..FFFFFF]
0k		Start Offset = 0ijkl (long word, 20 bits)
0l		Source data range end offset (from beginning of source wavesample data),
0m		Data offset = [0..FFFFFF]
0n		End Offset = 0mnop (long word, 20 bits)
0o		Destination data range start offset (from beginning of destination wavesample data),
0p		Data offset = [0..FFFFFF]
0q		Start Offset = 0qrst (long word, 20 bits)
0r		Clip prevention flag hi byte, (always 0)
0s		u = Clip prevention flag lo byte,
0t		clear = 00, set = 01
0u		

22	SCALE WAVEDATA	Scale the data based on a scaling ramp between the scale factor points.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
0e		Data range start offset (from beginning of wavesample data), Data offset = [0..FFFFFF]
0f		Start Offset = 0efgh (long word, 20 bits)
0g		
0h		
0i		Data range end offset (from beginning of wavesample data), Data offset = [0..FFFFFF]
0j		End Offset = 0ijkl (long word, 20 bits)
0k		
0l		
00		Scale factor start point. (stored as 2 bytes, hi = 8 bit integer, lo = 7 bit fraction), factor = mno. see note at top of section 4.4. start point = [0.0..255.99]
0m		
0n		
0o		
00		Scale factor end point. (stored as 2 bytes, hi = 8 bit integer, lo = 7 bit fraction), factor = pqr. see note at top of section 4.4. end point = [0.0..255.99]
0p		
0q		
0r		
00		Scale ramp depth hi byte (always 0)
0s		s = Scale ramp depth lo byte depth = [0..6] (0=3.0dB, 1=3.5db, ... 6=6.0dB)
23	INVERT WAVEDATA	Invert the wavesample data in the source range, i.e. negate the sample values.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
0e		Data range start offset (from beginning of wavesample data), Data offset = [0..FFFFFF]
0f		Start Offset = 0efgh (long word, 20 bits)
0g		
0h		
0i		Data range end offset (from beginning of wavesample data), Data offset = [0..FFFFFF]
0j		End Offset = 0ijkl (long word, 20 bits)
0k		
0l		

24	REVERSE WAVEDATA	Reverse the wavesample data in the source range.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, Layer # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
0e		Data range start offset (from beginning of wavesample data), Data offset = [0..FFFFFF]
0f		Start Offset = 0efgh (long word, 20 bits)
0g		
0h		
0i		Data range end offset (from beginning of wavesample data), Data offset = [0..FFFFFF]
0j		End Offset = 0ijkl (long word, 20 bits)
0k		
0l		
25	REPLICATE WAVEDATA	Append the wavesample data in the source range to itself repeatedly until the whole wavesample data block is filled.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
0e		Data range start offset (from beginning of wavesample data), Data offset = [0..FFFFFF]
0f		Start Offset = 0efgh (long word, 20 bits)
0g		
0h		
0i		Data range end offset (from beginning of wavesample data), Data offset = [0..FFFFFF]
0j		End Offset = 0ijkl (long word, 20 bits)
0k		
0l		

26	CROSS FADE LOOP	Make a cross fade loop of a certain size based on a scaling ramp.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
0e		Fade zone size (see note at top of section 4.4)
0f		Size = [0..FFFFFF], same as wavedata offset
0g		Fade zone = 0efgh (long word, 20 bits)
0h		
00		Scale ramp depth hi byte (always 0)
0i		i = Scale ramp depth lo byte depth = [0..6] (0=3.0dB, 1=3.5db, ... 6=6.0dB)
27	FADE IN WAVEDATA	Fade in the wavesample data in the source range.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
0e		Data range start offset (from beginning of wavesample data), Data offset = [0..FFFFFF]
0f		Start Offset = 0efgh (long word, 20 bits)
0g		
0h		
0i		Data range end offset (from beginning of wavesample data), Data offset = [0..FFFFFF]
0j		End Offset = 0ijkl (long word, 20 bits)
0k		
0l		
00		Scale ramp depth hi byte (always 0)
0m		m = Scale ramp depth lo byte depth = [0..6] (0=3.0dB, 1=3.5db, ... 6=6.0dB)

28	FADE OUT WAVEDATA	Fade out the wavesample data in the source range.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
0e		Data range start offset (from beginning of wavesample data), Data offset = [0..FFFFFF]
0f		Start Offset = 0efgh (long word, 20 bits)
0g		
0h		
0i		Data range end offset (from beginning of wavesample data), Data offset = [0..FFFFFF]
0j		End Offset = 0ijkl (long word, 20 bits)
0k		
0l		
00		Scale ramp depth hi byte (always 0)
0m		m = Scale ramp depth lo byte depth = [0..6] (0=3.0dB, 1=3.5db, ... 6=6.0dB)
29	REVERSE CROSS FADE LOOP	The whole loop section is faded out, reversed, added to itself, and then a normal cross fade loop is created.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
0e		Fade zone size (see note at top of section 4.4)
0f		Size = [0..FFFFFF], same as wavedata offset
0g		Fade zone = 0efgh (long word, 20 bits)
0h		
00		Scale ramp depth hi byte (always 0)
0i		i = Scale ramp depth lo byte depth = [0..6] (0=3.0dB, 1=3.5db, ... 6=6.0dB)

2A	ENSEMBLE CROSS FADE LOOP	A normal cross fade loop is created where the cross fade zone equals the loop length.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
00		Fade zone size (see note at top of section 4.4)
00		Fade zone = 00000 (long word, 20 bits)
00		
00		
00		Scale ramp depth hi byte (always 0)
0e		e = Scale ramp depth lo byte
		depth = [0..6] (0=3.0dB, 1=3.5db, ... 6=6.0dB)
2B	BOWTIE CROSS FADE LOOP	A cross fade loop is created such that the data at each of the loop points is the same. The drawing of the fade lines forms a bowtie shape, hence the name. The cross fade zone length must not be greater than half of the loop length.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
0e		Fade zone size (see note at top of section 4.4)
0f		Size = [0..FFFFFF], same as wavedata offset,
0g		Fade zone = 0efgh (long word, 20 bits)
0h		
00		Scale ramp depth hi byte (always 0)
0i		i = Scale ramp depth lo byte
		depth = [0..6] (0=3.0dB, 1=3.5db, ... 6=6.0dB)

2C	LENGTHEN LOOP	The data at the end of the loop is faded out, reversed, and added such that the faded sections overlap.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
0e		Fade zone size (see note at top of section 4.4)
0f		Size = [0..FFFFFF], same as wavedata offset,
0g		Fade zone = 0efgh (long word, 20 bits)
0h		
00		Scale ramp depth hi byte (always 0)
0i		i = Scale ramp depth lo byte
		depth = [0..6] (0=3.0dB, 1=3.5db, ... 6=6.0dB)
2D	MIX WAVEDATA	Mix the data of the source wavesample into the destination wavesample. If there is any pitch difference between the wavesamples, the command will fail because there is no way to correct the pitch difference.
00		Source instrument number hi byte (always 0)
0a		a = Source instrument number lo byte, # = [0..7]
00		Source layer number hi byte (always 0)
0b		b = Source layer number lo byte, # = [0..7]
0c		c = Source wavesample number hi byte
0d		d = Source wavesample number lo byte, # = [1..127]
00		Destination instrument number hi byte (=0)
0e		e = Destination instrument number lo byte
00		Destination layer number hi byte (always 0)
0f		f = Destination layer number lo byte, # = [0..7]
0g		g = Destination wavesample number hi byte
0h		h = Destination wavesample number lo byte
0i		i = Balance control hi byte (always 0)
0j		j = Balance control lo byte, # = [-127..127] (-127 is all source, 0 = equal balance, and 127 is all destination)

2E	MERGE/SPLICE WAVEDATA	Merge the data of the source wavesample into the destination wavesample using a variable cross-fade. A splice is achieved using a zero length cross-fade. If there is any pitch difference between the wavesamples, the command will fail because there is no way to correct the pitch difference.
00		Source instrument number hi byte (always 0)
0a		a = Source instrument number lo byte, # = [0..7]
00		Source layer number hi byte (always 0)
0b		b = Source layer number lo byte, # = [0..7]
0c		c = Source wavesample number hi byte
0d		d = Source wavesample number lo byte, # = [1..127]
00		Destination instrument number hi byte (=0)
0e		e = Destination instrument number lo byte
00		Destination layer number hi byte (always 0)
0f		f = Destination layer number lo byte, # = [0..7]
0g		g = Destination wavesample number hi byte
0h		h = Destination wavesample number lo byte
0i		Fade zone size [loop size > size < (loop start - wave-
0j		data start)],
0k		Size = [0..FFFFFF], same as wavedata offset
0l		Fade zone = 0ijkl (long word, 20 bits)
00		Scale ramp depth hi byte (always 0)
0m		m = Scale ramp depth lo byte depth = [0..6] (0=3.0dB, 1=3.5db, ... 6=6.0dB)
0n		n = Balance control hi byte
0o		o = Balance control lo byte, # = [-127..127] (-127 is all source, 0 = equal balance, and 127 is all destination)

2F	VOLUME SMOOTHING	The amplitude of a section of data is made the same as the amplitude at the beginning of the section. This is a functions like a digital dynamic compressor.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
0e		Data range start offset (from beginning of wavesample data), Data offset = [0..FFFFFF]
0f		Start Offset = 0efgh (long word, 20 bits)
0g		
0h		
0i		Data range end offset (from beginning of wavesample data), Data offset = [0..FFFFFF]
0j		End Offset = 0ijkl (long word, 20 bits)
0k		
0l		
00		Scale ramp depth hi byte (always 0)
0m		m = Scale ramp depth lo byte
		depth = [0..6] (0=3.0dB, 1=3.5db, ... 6=6.0dB)
00		Smoothness parameter hi byte (always 0)
0n		n = Smoothness lo byte, #=[0..4] (0 is very fine, 4 is very coarse)
41	NORMALIZE GAIN	The wavedata is scaled up to the full digital signal range.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]

42	SYNTHESIZED LOOP	Given the current loop points in the specified wave- sample and the smoothness parameter, a loop is gener- ated using an intelligent randomized algorithm.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
00		Smoothness parameter hi byte (always 0)
0e		e = Smoothness lo byte, #=[0..4] (0 is very fine, 4 is very coarse)
43	BIDIRECTIONAL CROSS FADE LOOP	This is a variation of the normal cross fade for bidirec- tional loops. The cross fade zone length must not be greater than half of the loop length.
00		Instrument number hi byte (always 0)
0a		a = Instrument number lo byte, # = [0..7]
00		Layer number hi byte (always 0)
0b		b = Layer number lo byte, # = [0..7]
0c		c = Wavesample number hi byte
0d		d = Wavesample number lo byte, # = [1..127]
0e		Fade zone size (see note at top of section 4.4)
0f		Size = [0..FFFFFF], same as wavedata offset,
0g		Fade zone = 0efgh (long word, 20 bits)
0h		
00		Scale ramp depth hi byte (always 0)
0i		i = Scale ramp depth lo byte depth = [0..6] (0=3.0dB, 1=3.5db, ... 6=6.0dB)

5 Command Status Codes

These codes are the low byte of the data word in a response command. Each code indicates how the receiver processed the preceding command or if it was able to process it at all. Each status code is sent in 12 bit word format as described in 2.3. The given meaning is intended as an explanation of the error and possible error recovery actions.

#	Name	Meaning
00	ACK	Yes, OK, it worked, cool, and it's jammin'.
01	WAIT	The receiver needs extra time to process the command, load an overlay, etc. The transmitter should reset its message timer for 30 seconds. Chill...
02	INSERT SYSTEM DISK	The system disk must be inserted in the EPS drive before the command is executed. The transmitter should prompt the user to insert the disk, and after the user is done, re-transmit the command.
03	INVALID PARAMETER NUMBER	A parameter number in the last GET or PUT parameter command doesn't exist.
04	INVALID PARAMETER VALUE	A parameter value in the last GET or PUT parameter command is out of range.
05	INVALID INSTRUMENT	The instrument(s) specified doesn't exist, i.e. is not loaded.
06	INVALID LAYER	The LAYER(s) specified doesn't exist.
07	LAYER IN USE	The destination LAYER already exists and cannot be used.
08	INVALID WAVE-SAMPLE	The wavesample(s) specified doesn't exist.
09	WAVESAMPLE IN USE	The destination wavesample already exists and cannot be used.
0A	INVALID WAVE-SAMPLE DATA RANGE	The data range specified doesn't exist within the specified wavesample.
0B	FILE NOT FOUND	The filename in the command cannot be found in the directory.
0C	MEMORY FULL	There is not enough RAM needed to execute the command.

0D	INSTRUMENT IN USE	The specified instrument cannot be used for the given command.
0E	NO MORE LAYERS	All eight layers are in use.
0F	NO MORE WAVESAMPLES	All 127 wavesamples are in use.
10	reserved	Reserved for internal EPS use.
11	WAVESAMPLE IS A COPY	The wavedata in the specified wavesample is a copy of data in another wavesample. The command can only be executed on the wavesample containing the data.
12	ZONE TOO BIG	The cross fade zone is too big. Make it smaller or move the loop points away from the ends of the wavesample data.
13	SEQUENCER MUST BE STOPPED	The command could not be executed because the sequencer was running. Stop the sequencer and redo the command.
14	DISK ACCESS IN PROGRESS	Current disk activity prevented the execution of the command.
15	DISK FULL	The disk is full and no files can be stored.
16	LOOP IS TOO LONG	The processing command could not be completed because the loop length is too long. This is currently used only by the synthesized loop command.
17	NAK	Something was wrong with the last data transfer which could not be processed.
18	NO LAYER EDIT	When the wavesample number is zero, a PUT PARAMETER message will effect all wavesamples on the given layer. This status code is used when this type of editing is not allowed, especially wavedata offset editing.
19	NO MORE PITCH TABLES	Only eight pitch tables can be created within an instrument.
1A	CROSS FADE LENGTH IS ZERO	The cross fade length cannot be zero in this command.

1B	CROSS FADE LENGTH IS GREATER THAN 50%	The cross fade zone must be less than or equal to fifty percent of the loop length.
1C	LOOP START TOO CLOSE TO SAMPLE START	The processing command could not be completed because there was not enough room between sample start and loop start.
1D	LOOP END TOO CLOSE TO SAMPLE END	The processing command could not be completed because there was not enough room between loop end and sample end.
1E	QUIET LAYER	The layer being edited is not in the patch so any parameter changes will not be heard.

6 Button Numbers

The following table lists the valid key numbers for use in the VIRTUAL BUTTON PRESS command.

Logical Key Number	Front Panel Button Name
0	Instrument 1
1	Instrument 2
2	Instrument 3
3	Instrument 4
4	Instrument 5
5	Instrument 6
6	Instrument 7
7	Instrument 8
16	Sample
17	Command Mode
18	Key Range (instrument or wavesample)
19	Edit Mode
20	Load Mode
21	MIDI
22	Seq • Song
23	Instrument
24	System

Logical Key Number	Front Panel Button Name
32	Up Arrow (increment)
33	Down Arrow (decrement)
34	Left Arrow (scroll left)
35	Cancel • No
36	Right Arrow (scroll right)
37	Enter • Yes
48	0 • Track
49	1 • Env 1
50	2 • Env 2
51	3 • Env 3
52	4 • Pitch
53	5 • Filter
54	6 • Amp
55	7 • LFO
56	8 • Wave
57	9 • Layer

7 Parameter Block Data Descriptions

The parameter blocks transmitted in the PUT INSTRUMENT, PUT LAYER, PUT WAVESAMPLE, and PUT PITCH TABLE commands are described here. See section 4.2 for detailed descriptions of the commands.

7.1 Instrument Parameter Block

This table describes the data transmitted by the second message of the PUT INSTRUMENT command. All data in the data block is transmitted using the 16 bit word format described in section 2.3. The table describes the data as it exists in RAM, i.e. the given words offsets are the RAM offsets not the MIDI data offsets. Refer to section 9 for the parameter value ranges.

Word Offset <i>decimal</i>	Data Word Description <i>(all values in hi byte of word)</i>
00-11	Name - 12 ASCII bytes, one byte per word
12	MIDI Channel - (<i>outbound</i>)
13	MIDI Program Number - (<i>outbound</i>)
14	MIDI Pressure - (<i>outbound</i>)
15	Total Instrument Size in Blocks (1 Block = 256 words)
16	Key Destination (LOCAL, MIDI, or BOTH)
17	Patch 0 (bit map of layers, bit 0=LYR 1, bit 1=LYR 2, ...)
18	Patch 1 (bit map of layers, bit 0=LYR 1, bit 1=LYR 2, ...)
19	Patch 2 (bit map of layers, bit 0=LYR 1, bit 1=LYR 2, ...)
20	Patch 3 (bit map of layers, bit 0=LYR 1, bit 1=LYR 2, ...)
21	Key Down Layers (bit map of layers, bit 0=LYR 1, bit 1=LYR 2, ...)
22	Key Up Layers (bit map of layers, bit 0=LYR 1, bit 1=LYR 2, ...)
23-25	unused
26	Key Range - Lo Key
27	Key Range - Hi Key
28	Transposition - number of semitones (signed)
29-44	Pitch Table Offsets - (<i>relative to instrument base address in EPS RAM</i>)
45-60	Layer Offsets - (<i>relative to instrument base address in EPS RAM</i>)
61-316	Wavesample Offsets - (<i>relative to instrument base address in EPS RAM</i>)
317-322	unused

7.2 Layer Parameter Block

This table describes the data transmitted by the second message of the PUT LAYER command. All data in the data block is transmitted using the 16 bit word format described in section 2.3. The table describes the data as it exists in RAM, i.e. the given words offsets are the RAM offsets not the MIDI data offsets. Refer to section 9 for the parameter value ranges.

Word Offset <i>decimal</i>	Data Word Description <i>(all values in hi byte of word)</i>
00-11	Name - 12 ASCII bytes, one byte per word
12	Glide Mode
13	Glide Time
14	Legato Layer Number
15	Velocity Lo
16	Velocity Hi
17	Pitch Table Number
18	unused
19-106	Layer Map - <i>(one wavesample number per key in hi byte)</i>

7.3 Wavesample Parameter Block

This table describes the data transmitted by the second message of the PUT WAVESAMPLE command. All data in the data block is transmitted using the 16 bit word format described in section 2.3. The table describes the data as it exists in RAM, i.e. the given words offsets are the RAM offsets not the MIDI data offsets. Refer to section 9 for the parameter value ranges.

Word Offset <i>decimal</i>	Data Word Description <i>(all values in hi byte of word)</i>
00-11	Name - 12 ASCII bytes, one byte per word
12	Wavesample Copy Number - if non-zero, this is the number of the wavesample containing the sample data
13	Wavesample Copy Layer - if non-zero, this is the number of the layer containing the original wavesample
14-35	Pitch Envelope #1 - see section 7.3.1
36-57	Filter Envelope #2 - see section 7.3.1
58-79	Amplitude Envelope #3 - see section 7.3.1
80	Root Key - <i>MIDI key number</i>
81	Pitch Envelope Amount
82	LFO Amount
83	Random Modulation Amount
84	Pitch Wheel Bend Range
85	Modulation Source
86	Fine Tune - <i>signed 7 bit fraction in hi byte</i>
87	Modulation Amount
88	Filter Mode
89	FC #1 Cutoff
90	FC #2 Cutoff
91	FC #1 Keyboard Amount
92	FC #2 Keyboard Amount
93	FC #1 Filter Envelope Amount
94	FC #2 Filter Envelope Amount
95	FC #1 Modulation Source
96	FC #2 Modulation Source
97	FC #1 Modulation Amount
98	FC #2 Modulation Amount
99	Volume
100	Amplitude Modulation Source
101	Amplitude Crossfade Curve Point A
102	Amplitude Crossfade Curve Point B
103	Amplitude Crossfade Curve Point C
104	Amplitude Crossfade Curve Point D

105	Pan Position - <i>including separate out assignment</i>
106	Amplitude Modulation Amount
107	LFO Waveform
108	LFO Speed
109	LFO Depth
110	LFO Delay Time
111	LFO Modulation Source
112	LFO Mode
113	Random Modulator Frequency
114	Loop Mode
115-118	Sample Start Offset (<i>left justified 32 bit field using hi bytes of each word - shift right 9 for word offset</i>)
119-122	Sample End Offset (<i>left justified 32 bit field using hi bytes of each word - shift right 9 for word offset</i>)
123-126	Loop Start Offset (<i>left justified 32 bit field using hi bytes of each word - shift right 9 for word offset</i>)
127-130	Loop End Offset (<i>left justified 32 bit field using hi bytes of each word - shift right 5 for 4 bit loop end fraction</i>)
131	Sample Rate - <i>sample period = rate * 1.6 microseconds</i>
132	Key Range - Lo Key
133	Key Range - Hi Key
134	Start or Loop Modulation Source
135	Start or Loop Modulation Amount
136	Start or Loop Modulation Range
137	Modulation Type (<i>none, sample start, loop, or both</i>)
138	unused

7.3.1 Wavesample Envelope Description

This table describes the structure of each of the three envelopes in each wavesample. All offsets are given from the base of the envelope, e.g. the base of the filter envelope is offset 36 words from the start of the data block in a PUT WAVESAMPLE command. Refer to section 9 for the parameter value ranges.

Word Offset <i>decimal</i>	Data Word Description <i>(all values in hi byte of word)</i>
0	Envelope Type (<i>default envelopes</i>)
1	Soft Level 0 - initial level
2	Hard Level 0 - initial level
3	Time 1 - attack time; time from initial level to level 1
4	Soft Level 1 - peak level
5	Hard Level 1 - peak level
6	Time 2 - first decay time; time from level 1 to level 2
7	Soft Level 2
8	Hard Level 2
9	Time 3 - second decay; time from level 2 to level 3
10	Soft Level 3
11	Hard Level 3
12	Time 4 - third decay; time from level 3 to level 4
13	Soft Level 4 - sustain
14	Hard Level 4 - sustain
15	Time 5 - release time; time from level 4 to level 5
16	Velocity Switch - soft level on/off
17	Level 5 - release breakpoint relative to sustain level (+/-)
18	Time 6 - second release time; time from level 5 to 0
19	Time 1 velocity sensitivity
20	Keyboard Time Scaling
21	Mode (0=normal, 1=cycle, 2=repeat)

7.4 Pitch Table Block

This table describes the data transmitted by the second message of the PUT PITCH TABLE command. All data in the data block is transmitted using the 16 bit word format described in section 2.3. The table describes the data as it exists in RAM, i.e. the words offsets given are the RAM offsets not the MIDI data offsets.

Word Offset <i>decimal</i>	Data Word Description <i>(all values in hi byte of word)</i>
00-11	Name - 12 ASCII bytes, one byte per word
12-99	Key Map - 88 key values, one key per word <i>(for each word, bits 15:9=semitone, and bits 8:3=semitone fraction)</i>
100-106	unused

8 Example of a Wavedata Transfer

Assume that EXT wants to receive a portion of wavesample data from the EPS. This is one of the most common data transfers, and the following steps describe the commands sent to and from the EPS to affect the data transfer.

1. to EPS *from* EXT:

GET WAVESAMPLE DATA: specify the instrument, layer, wavesample, and the start and end offsets.

2. to EXT *from* EPS:

RESPONSE: if all the parameters of the GET WAVESAMPLE DATA command were OK, then the ACK status code is used, otherwise an error status code is used and the data transfer is complete.

3. to EXT *from* EPS:

PUT WAVESAMPLE DATA (first part): the parameters here are the same as the GET WAVESAMPLE DATA command.

4. to EPS *from* EXT:

RESPONSE: if EXT still wants the data, the ACK status code should be used, otherwise the data transfer is complete.

5. to EXT *from* EPS:

PUT WAVESAMPLE DATA (second part): all the samples within the data range given in the GET WAVESAMPLE DATA are transmitted using the 16 bit word format described in section 2.3. The block of samples forms the message within the system exclusive packet according to section 2.

6. to EPS *from* EXT:

RESPONSE: if the data was successfully received, the ACK status code should be used. If an error occurred during the second part of the PUT WAVESAMPLE DATA command, the data transfer must be started again from step 1 since the EPS will not retransmit the second part of the PUT WAVESAMPLE DATA again regardless of the status code.

9 Parameter Numbers

The following table of parameter numbers is used in the GET and PUT PARAMETER commands. The complete parameter number word consists of two bytes: the page number (high byte) and the item number (low byte). The number is transmitted using the 12 bit word format described in section 2.3. An easy way to remember parameter numbers in a MIDI data dump is high byte times four followed by the low byte. For example, Free System Blocks is transmitted via MIDI bytes \$34 \$00 not \$0D \$00. Parameter values are right justified within a 24 bit binary word and transmitted as two 12 bit words according to section 2.3.

9.1 SYSTEM PARAMETERS

SYSEX HIGH BYTE: 13

	LOW BYTE	RANGE
Free System Blocks	0	0-10000 (Read Only)
Free Disk Blocks	1	0-10000 (Read Only)
Master Tune	2	-127 to +127
Global Bend Range	3	0-12
Touch Sensitivity	4	0-15 (Table 1)
Mod Pedal Mode	5	0-1 (Volume-Modulator)
Sustain Pedal Mode	6	0-1 (Sustain-Left Patch Select)
Aux Pedal Mode	7	0-1 (Start/Stop-Right Patch Select)
Autoloop Switch	8	0-1 (Off-On)

9.2 MIDI PARAMETERS

SYSEX HIGH BYTE: 12

	LOW BYTE	RANGE
Base Channel	0	0-15 (Channels 1-16)
Midi Transmit Mode	1	0-1 (Base or Instrument Channels)
Base Channel Pressure Mode	2	0-2 (Off-Key-Channel)
Midi In Mode	3	0-4 (Omni-Poly-Multi-Mono A-Mono B)
Midi Controllers Enable	4	0-1 (Off-On)
Midi SYSEX Enable	5	0-1 (Off-On)
Midi Program Change Enable	6	0-1 (Off-On)
Midi Song Position Enable	7	0-1 (Off-On)
Midi XCTRL Value	8	0-127

9.3 INSTRUMENT PARAMETERS

SYSEX HIGH BYTE: 10

	LOW BYTE	RANGE
Patch	0	0-255
Key Down Layers	1	0-255
Key Up Layers	2	0-255

Description: The above values select the layers in each parameter. Each layer is represented by a bit; if the bit is set, that layer will sound.

Midi Channel	3	0-15 (Channels 1-16)
Midi Program	4	1-127
Pressure Mode	5	0-2 (Off-Key-Channel)
Instrument Size	7	0-10000 (Read Only)
Send Keys To	6	0-2 (Both-Local-Midi)
Range Low Key	10	0-127
Range High Key	11	0-127
Transpose Amount-Octave	12	0-5
Transpose Amount-Semitone	13	0-12

9.4 LAYER PARAMETERS

SYSEX HIGH BYTE: 9

	LOW BYTE	RANGE
Glide Mode	0	0-2 (Off-Mono-Pedal)
Glide Time	1	0-127
Legato Layer	2	0-7
Velocity Low	3	0-127
Velocity High	10	0-127
Pitch Table	4	0-7

9.5 WAVESAMPLE PARAMETERS

SYSEX HIGH BYTE: 8

	LOW BYTE	RANGE
Loop Mode	0	0-4 (Table 8)
Loop Mod Type	6	0-3 (Off-Loop-Start-Both)
Loop Mod Source	7	0-17 (Table 2)
Loop Mod Amount 1	8	0-127
Loop Mod Amount 2	9	0-21 (Table 7)
Wavedata Start	21	0-FFFFFFh
Wavedata End	22	0-FFFFFFh
Loop Start	23	0-FFFFFFh
Loop End	24	0-FFFFFFh

Loop End Fractional	10	0-127
Key Range Low	11	0-127
Key Range High	12	0-127
Loop Position	25	0-FFFFFFh
Sample Rate	13	0-127

9.6 ENVELOPE PARAMETERS

SYSEX HIGH BYTE: 1-2-3 (ENV1 = 1, ENV2 = 2, ENV3 = 3)

	LOW BYTE	RANGE
Envelope Type	0	0-15 (Table 3)
Level 0 Hard	1	0-127
Level 0 Soft	2	0-127
Level 1 Hard	11	0-127
Level 1 Soft	15	0-127
Level 2 Hard	12	0-127
Level 2 Soft	16	0-127
Level 3 Hard	13	0-127
Level 3 Soft	17	0-127
Level 4 Hard	14	0-127
Level 4 Soft	18	0-127
Time 1	3	0-99
Time 2	19	0-99
Time 3	20	0-99
Time 4	21	0-99
Time 5	22	0-99
2nd Release Time	4	0-127
2nd Release Level	23	-127 to +127
Attack Time Velocity	5	0-99
Keyboard Scaling	6	0-127
Soft Velocity On/Off	7	0-1 (Off-On)
Envelope Mode	8	0-2 (Normal-Cycle-Repeat)

9.7 PITCH PARAMETERS

SYSEX HIGH BYTE: 4

	LOW BYTE	RANGE
Root Key	1	0-127
Fine Tune	10	-127 to +127
LFO Amount	2	0-127
Envelope 1 Amount	3	-127 to +127
Random Frequency	5	0-127
Random Amount	12	-127 to +127

Bend Range	6	0-13 (13=global)
Modulation Source	7	0-18 (Table 2)
Modulation Amount	11	-127 to +127

9.8 FILTER PARAMETERS

SYSEX HIGH BYTE: 5

	LOW BYTE	RANGE
Filter Mode	0	0-3 (Table 4)
Filter 1 Cutoff	1	0-127
Filter 2 Cutoff	11	0-127
Filter 1 Envelope 2 Amount	2	-127 to +127
Filter 2 Envelope 2 Amount	12	-127 to +127
Filter 1 Keyboard Amount	3	-127 to +127
Filter 2 Keyboard Amount	13	-127 to +127
Filter 1 Modulation Source	7	0-18 (Table 2)
Filter 1 Modulation Amount	14	-127 to +127
Filter 2 Modulation Source	8	0-18 (Table 2)
Filter 2 Modulation Amount	15	-127 to +127

9.9 VOLUME PARAMETERS

SYSEX HIGH BYTE: 6

	LOW BYTE	RANGE
Wavesample Volume	1	0-127
Pan	2	0-17 (Table 5)
Modulation A	3	0-127
Modulation B	11	0-127
Modulation C	4	0-127
Modulation D	12	0-127
Modulation Source	7	0-18 (Table 2)
Modulation Amount	10	0-127

9.10 LFO PARAMETERS

SYSEX HIGH BYTE: 7

	LOW BYTE	RANGE
LFO Wave	1	0-6 (Table 6)
LFO Speed	2	0-99
LFO Depth	3	0-127
LFO Delay	4	0-99
LFO Mode	5	0-2 (Reset Off-Reset On-Humanize)
LFO Modulation Source	7	0-18 (Table 2)

9.11 EDIT PARAMETERS

SYSEX HIGH BYTE: 14

	LOW BYTE	RANGE
Edit Instrument	0	0-7 (Instruments 1-8)
Edit Layer	1	0-7 (Layers 1-8)
Edit Wavesample	2	0-128 (0=ALL)

PARAMETER TABLES

TABLE 1

0 = SOFT 1
1 = SOFT 2
2 = SOFT 3
3 = SOFT 4
4 = MEDIUM 1
5 = MEDIUM 2
6 = MEDIUM 3
7 = MEDIUM 4
8 = FIRM 1
9 = FIRM 2
10 = FIRM 3
11 = FIRM 4
12 = HARD 1
13 = HARD 2
14 = HARD 3
15 = HARD 4

TABLE 2

0 = LFO
1 = RANDM
2 = ENV1
3 = ENV2
4 = PR+VL
5 = VEL
6 = VEL 1
7 = VEL 2
8 = KBD
9 = PITCH
10 = WHEEL
11 = PEDAL
12 = XCTRL
13 = PRESS
14 = WL+PR
15 = OFF

TABLE 3

0 = CURRENT VALUE
1 = FULL ON
2 = ALL ZEROS
3 = FULL VELRANGE
4 = SLOW STRING
5 = PIANO DECAY

6 = PERCUSSION
 7 = RAMP UP
 8 = RAMP DOWN
 9 = SHORT BLIP
 10 = BRASS FILTER
 11 = REPEAT TRIANG
 12 = REPEAT RAMP
 13 = WIND DRIVEN
 14 = REVERB
 15 = SAVED

TABLE 4

0 = F1=2/LP F2=2/HP
 1 = F1=3/LP F2=1/HP
 2 = F1=2/LP F2=2/LP
 3 = F1=3/LP F2=1/LP

TABLE 5

0 = WAVESAMPLE
 1 = *-----
 2 = -*-----
 3 = --*-----
 4 = ---*-----
 5 = ----*-----
 6 = -----*--
 7 = -----*-
 8 = -----*
 9 = SOLO OUT 1
 10 = SOLO OUT 2
 11 = SOLO OUT 3
 12 = SOLO OUT 4
 13 = SOLO OUT 5
 14 = SOLO OUT 6
 15 = SOLO OUT 7
 16 = SOLO OUT 8
 17 = RANDOM PAN
 18 = KEYBOARD

TABLE 6

0 = TRIANGLE
 1 = SIN/TRIANGLE
 2 = SIN WAVE
 3 = POS/TRIANGLE

4 = POS/SIN WAVE
5 = SAWTOOTH
6 = SQUARE

TABLE 7

0 = 2 MG
1 = 1 MG
2 = 512 K
3 = 256 K
4 = 128 K
5 = 64 K
6 = 32 K
7 = 16 K
8 = 8 K
9 = 4 K
10 = 2 K
11 = 1 K
12 = 512 B
13 = 256 B
14 = 128 B
15 = 64 B
16 = 32 B
17 = 16 B
18 = 8 B
19 = 4 B
20 = 2 B
21 = 1 B

TABLE 8

0 = FORWARD-NO LOOP
1 = BACKWARD-NO LOOP
2 = LOOP FORWARD
3 = LOOP BIDIRECTION
4 = LOOP AND RELEASE

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